1. Aim of the practical: Partition the disk to efficiently allocate and manage storage space.

2. Objective: Partition the disk to efficiently allocate and manage storage space.

3. Hardware and Software Requirements:

a) **Disk Drive**: Physical disk (HDD/SSD) with enough space.
   Computer System: Host system running virtualization software.
b) **Operating System**: OS on the virtual machine (e.g., Fedora).

4. Commands/Steps:

Step-1: lsblk:- Shows a list of all block devices, including discs and partitions, along with the mount points, sizes, and kinds of each.

```
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   25G  0 disk
├─sda1   8:1    0    1M  0 part
├─sda2   8:2    0    1G  0 part /boot
└─sda3   8:3    0   24G  0 part /home
                                /
sr0     11:0    1 1024M  0 rom
zram0  252:0    0  1.9G  0 disk [SWAP]
```

Step-2: sudo parted /dev/sda:- Launches the parted tool to manage partitions on the specified disk.

```
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.
```

Step-3: It will ask for the password, and will prompt welcome message.

```
For security reasons, the password you type will not be visible.

[sudo] password for varunarora:
GNU Parted 3.6
Using /dev/sda
Welcome to GNU Parted! Type 'help' to view a list of commands.
```

Step-4: Print Displays the current partition table of the disk, showing all partitions and their sizes. This helps in identifying which partition you want to resize.

```
(parted) print
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sda: 26.8GB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags: pmbr_boot

Number  Start   End     Size    File system  Name  Flags
 1      1049kB  2097kB  1049kB                     bios_grub
 2      2097kB  1076MB  1074MB  ext4               bls_boot
 3      1076MB  26.8GB  25.8GB  btrfs
```

Step-5: Resizes the specified partition to end at the new size. Replace [partition number] (we are writing 3 as we want to edit the 3rd disk space with 10gb space).

```
(parted) resizepart 3 10GB
Warning: Partition /dev/sda3 is being used. Are you sure you want to continue?
Yes/No? yes
Warning: Shrinking a partition can cause data loss, are you sure you want to
continue?
Yes/No? yes
```

Step-6: Creates a new primary ext4 partition starting at 10GB and using all remaining disk space.

```
(parted) mkpart primary ext4 10GB 100%
(parted) quit
Information: You may need to update /etc/fstab.
```

5. Result:

Now we have done partitioning and we have created sda4.

lsblk:

```
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   25G  0 disk
├─sda1   8:1    0    1M  0 part
├─sda2   8:2    0    1G  0 part /boot
├─sda3   8:3    0  8.3G  0 part /home
│                               /
└─sda4   8:4    0 15.7G  0 part
sr0     11:0    1 1024M  0 rom
zram0  252:0    0  1.9G  0 disk [SWAP]
```

6. Learning outcomes (What I have learned):

a) Learned about Linux command.
b) Learned about Partition.

Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet		8 Marks
2.	Viva		10 Marks
3.	Simulation		12 Marks
	Total		30 Marks

Teacher Signature